

WASP-43b

R-0615

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180303 · created 2026-07-25T10:18:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458180.7966 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.024
Transit depth (Rp/Rs)^2	2.78 +/- 0.81 [%]
Orbital Inclination (inc)	81.2 +/- 2.24
Airmass coefficient 1 (a1)	0.844 +/- 0.013
Airmass coefficient 2 (a2)	0.117 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#3 - [382, 55]
Optimal Method	PSF photometry
Transit Duration (day)	0.046 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.167 ± 0.024 vs 0.1594 (deviation 0.32σ)
Duration fit vs published	0.046 d vs 0.0512 d (deviation 0.48σ)
Mid-time O–C	3.5 min
Depth SNR	7.0
Residual scatter	1.55 %

Light curve

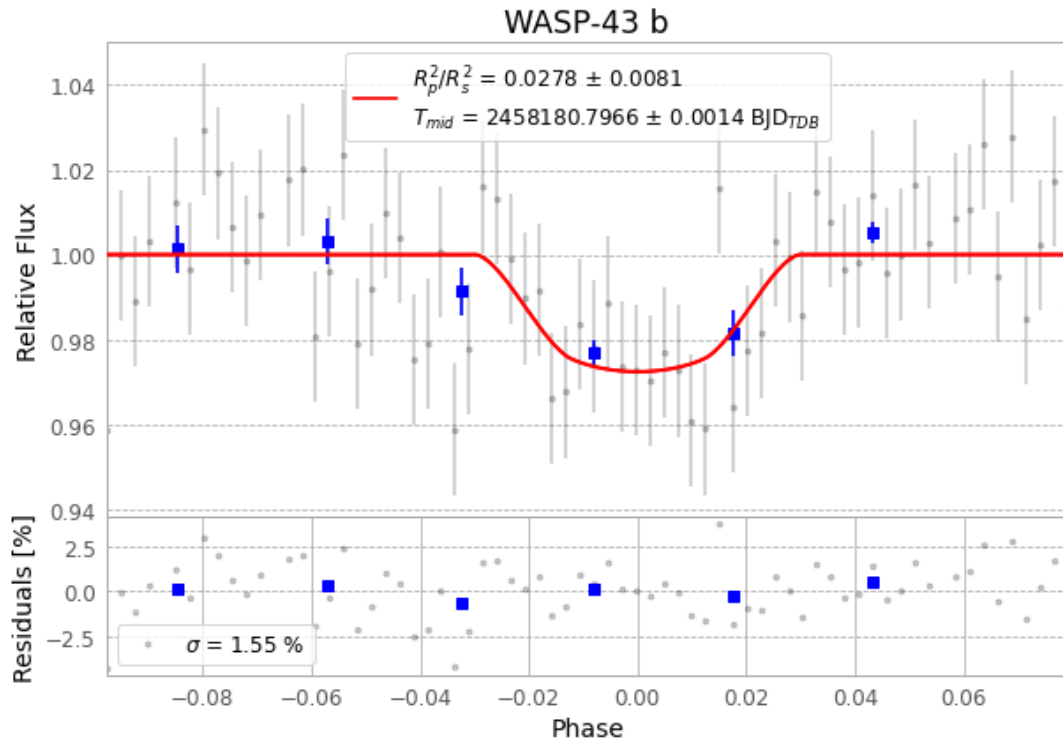


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-02.png (SHA-256 545ddc43daa7afa3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [459, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cc7dba801c983df7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), WASP-43180303050312.FITS → WASP-43180303083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180303.log (c5a0f63f196b...), FinalLightCurve_WASP-43 b_2018-03-02.png (545ddc43daa7...), FinalLightCurve_WASP-43 b_2018-03-02.csv (95e02946b77f...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 10:18Z.